

Laurentide SynthWorks

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VG2 - PASSIVE DUAL VACTROL GATE



VG2 is a Dual Passive Vactrol Gate. No power needed! Each Vactrol Gate has 1 CV Input (gate, envelope, LFO, etc.), 1 Audio Input and 1 Audio Output. The CV Input hits the Vactrol LED, allowing audio to pass through. In addition, there is a simple Low Pass Filter applied to the Audio Output with an Amount (Cutoff) knob. As this is a passive circuit, it will not give you full cutoff control, but it does allow the taming of harsh waveshapes. The LPF is simply the same concept as a guitar's "tone" control. Also, there is an LED at the CV Input for quick visual reference and LED light action!

One thing to keep in mind with a passive circuit using a vactrol is there may be varying differences between each vactrol. One may be slightly "louder" than the other, or have a longer release time or shorter release. These are variations that are inherent with vactrols and can usually be adjusted with a powered circuit, but not with a passive circuit. But, this adds a bit of variation to each vactrol gate making each output unique. It may be found that one output is better for percussive sounds and one is better for melodic sequences. There may also be no noticeable variations at all, but this is part of the vactrol charm and why we love them so much! Each Vactrol that is supplied with the kits are hand-selected by Laurentide SynthWorks. The discrepancies are bleed when silent & top output level. We try to pick vactrol's that have little or no bleed when no CV is applied and that they all have a similar output level, so you won't have to worry about one being waay quieter than the other.

As this is a passive circuit, it is advisable to use a Buffered Multiple before the CV Input to maintain voltage/amplitude, especially if you are using the top CV to drive both gates..

4hp
+12V 0mA
-12V 0mA

BUILD DOCUMENTATION

BOM

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BUILD

As these are photoresistors and LED's, This is a good kit for beginners. There's no voltage required so no chances of frying anything out. The build is pretty straight forward. Soldering is best done in this order:

1. Resistors (they don't have a polarity so can go in either way)



4k7 Resistor



33k Resistor



1k Resistor

The optional 1k resistor is to alleviate any audio bleed when there is no voltage at the CV IN. When the capacitor is installed I did not notice any or very little bleed. It does help though when the LPF is fully opened. There is a slight but noticeable high frequency drop when the resistor is installed. Try it first without the 1k. If you notice any audio bleed that is more than you'd like, install the 1k and test again. If you find you liked it before installing the 1k, just clip it out. No need to desolder.

The 4.7k resistor is for the Vactrol LED. I originally started with a 1k but found some modules (Make Noise Maths & Functions) would not work. The current pull from the Vactrol LED was too much. If you find a module doesn't jive, try increasing the resistor value. Maybe start with 5k, 10k etc. Note though that the higher the resistor the less light will be emitted by the internal LED.

2. Capacitors (they don't have a polarity so can go in either way)



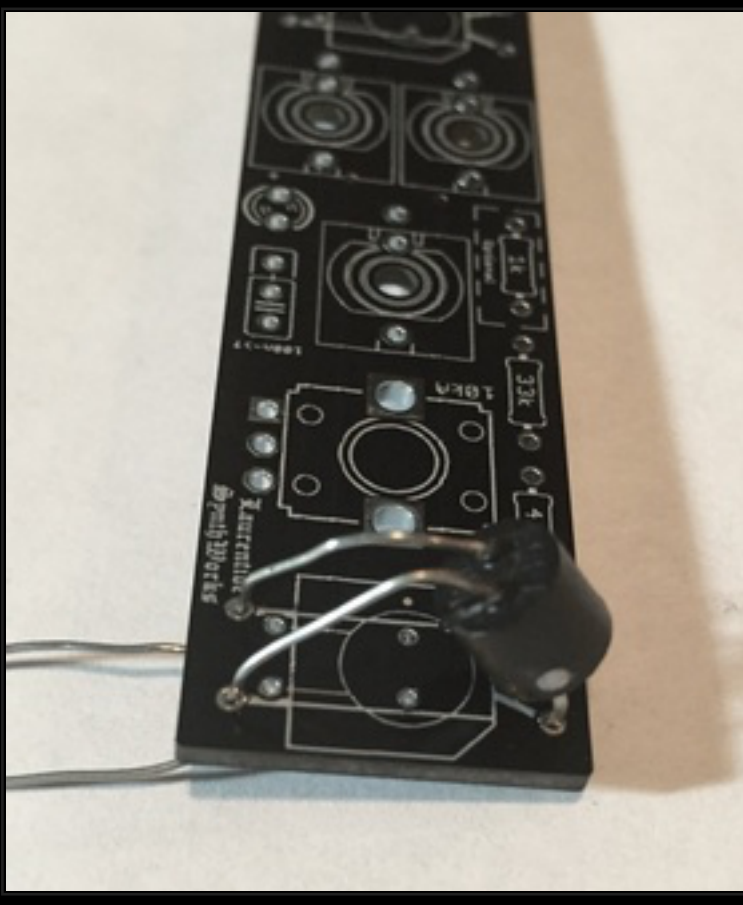
Any MLCC (multi-layer ceramic capacitor), film, polystyrene, etc. capacitor starting from 100n (.1u) on up can be used (the kits come with 330n film caps). The higher the value, the more high frequencies will be cut off. We don't need to debate which vintage, orange drop fancy ass capacitor sounds better than the other. This isn't a Strat

3. Vactrols

When installing the vactrols, notice that there are 2 different package types over the same footprint. Pay attention to the orientation. The ones supplied in the kit have a white dot next to the LED - (cathode). Line this up with the dot on the PCB then bend the photoresistor legs into the other holes.



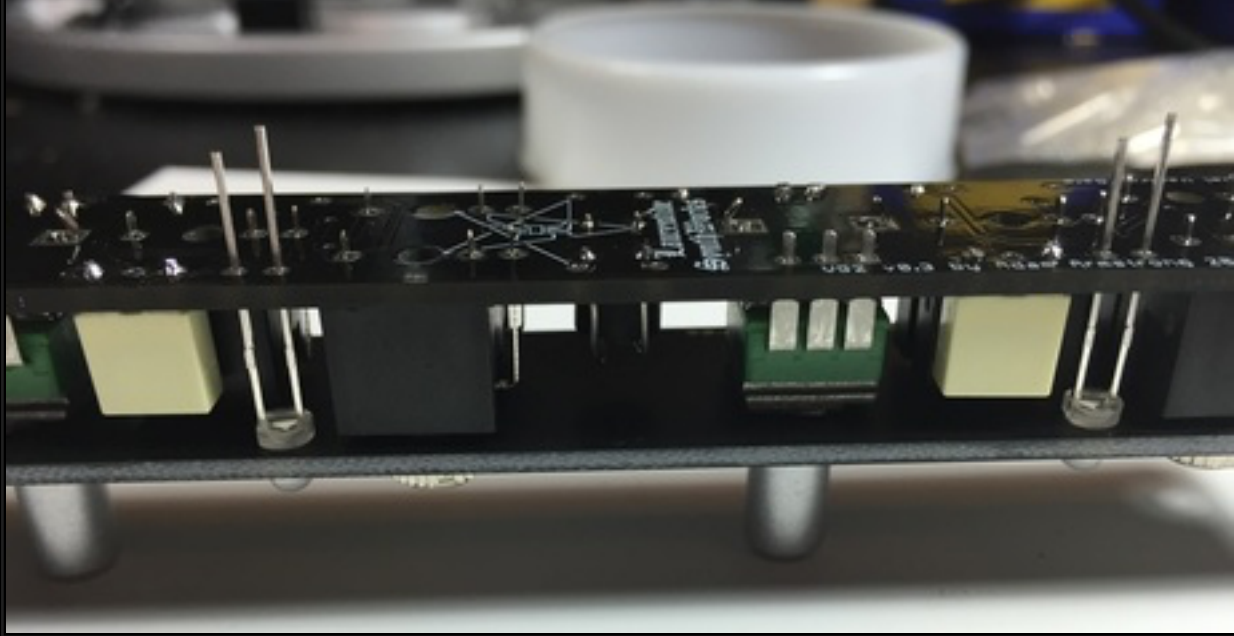
NOTE: The vactrols that are supplied in the kits seem to have varying sized legs on the cathode side of the LED (the white dot side) and it won't fit in the hole. You can either take a pair of needlenose pliers and squeeze the leg a bit to thin it out, or use the other package holes instead. Keep in mind that the LED cathode and anode are reversed, so the white dot will line up with the angled side.



Once all of these parts are soldered in, add all of the jacks, potentiometers and LED's. **NOTE:** don't forget to cut the guide pin on the potentiometer. If this is left, the panel won't sit flat across the pots and jacks.



Then, put the panel on top and secure with a couple of the nuts. Push the LED through the hole on the panel to desired height, or all the way.



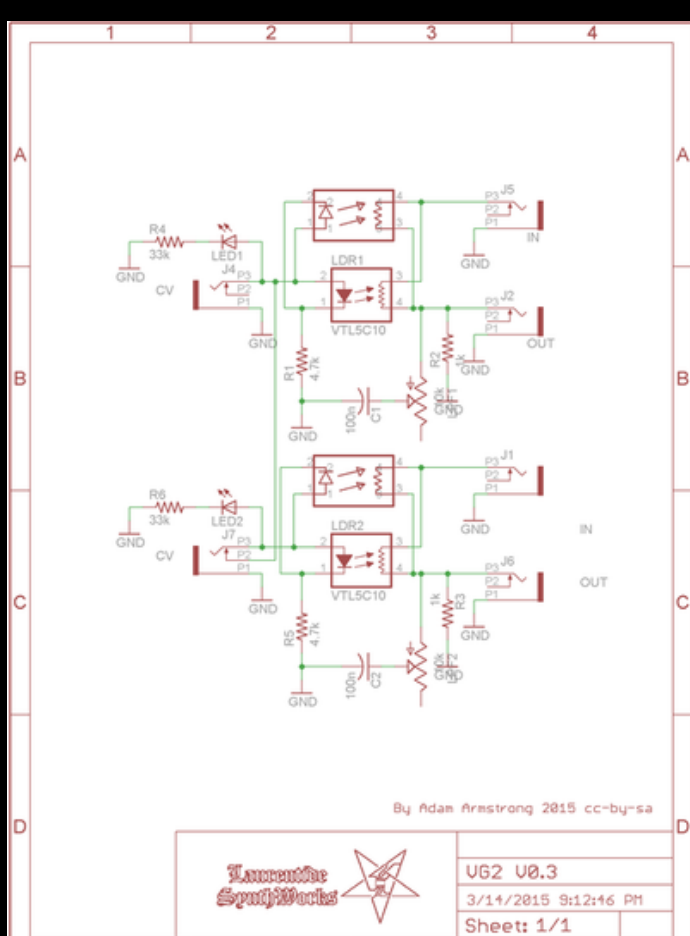
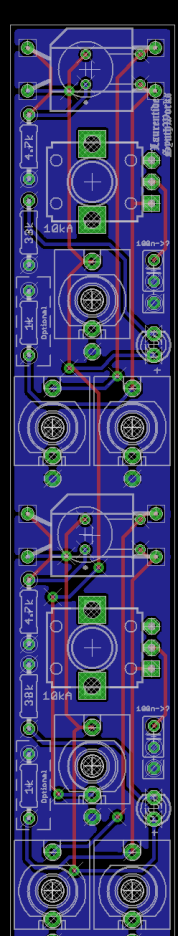
Once everything is secure, flip it over and only solder 1 lead of each part. Then, flip back and make sure everything is lined up. If everything looks good, solder the remaining leads/legs. Add the remaining nuts and knobs to the pots.

That's it! Now send it a Gate, route a VCO to the IN and listen to the OUT's nice vactrol gateage..

PCB MOD

One of the features is whatever is sent to the CV jack on the top gate is also sent to the CV jack on the bottom gate. The downside to this is there is a slight voltage drop causing the both outputs to be slightly quieter than if the CV was only driving 1 gate. If you are only using one the inputs and have something against using the bottom gate, you can plug a dummy cable into the bottom CV, or completely disable it by cutting the trace. You will want to cut the trace between the top CV Jack top most pin and the bottom CV jacks middle pint (just above the GND pin). Double check continuity 1st before cutting just to be safe. You can see the connection to cut on the Schematic below. Top CV Jack P3 to bottom CV jack P2. Be sure to cut the trace after the via just below the top CV. This sends the incoming CV to the status LED.

REFERENCE PCB & SCHEMATIC



PCB & PANEL FILES

The PCB & Panel were designed with Eagle Cadsoft. A limited version can be downloaded for free from their website.

[vg2_pcb_v2.zip](#)
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